DECODELABS: DATA ANALYTICS INTERNSHIP

Project 2: Exploratory Data Analysis (EDA)

EDA Report

Intern Name: Ravi kishan
Date Submitted: 09 july 2026
Dataset: Cleaned Dataset for Data Analytics (DecodeLabs)
Tool Used: Microsoft Excel

Problem Statement

Analyze 1,200 order records from a DecodeLabs e-commerce business spanning January 2023 to June 2025 to identify revenue trends, product performance, customer behavior patterns and operational inefficiencies that can drive smarter business decisions.

Methodology

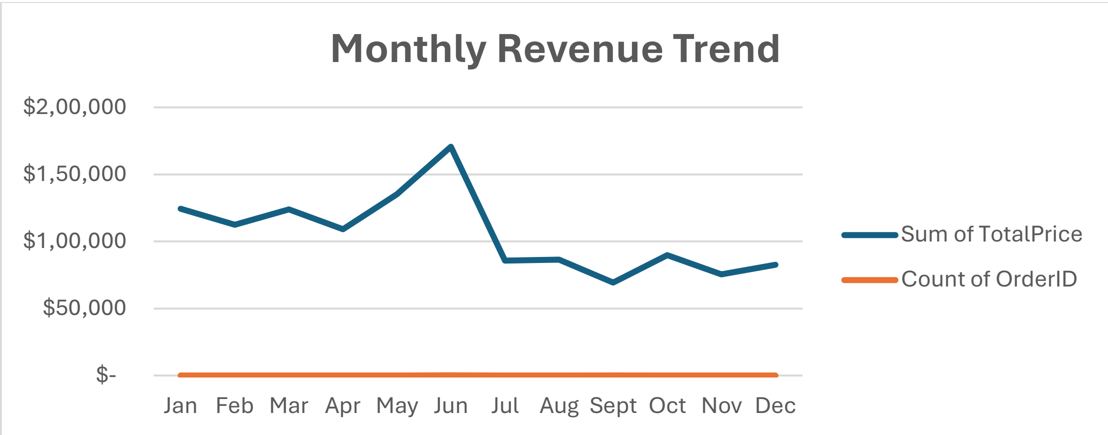
The following EDA techniques were applied using Microsoft Excel:

- Descriptive Statistics: Mean, Median, Mode, Min, Max and Standard Deviation calculated for all numeric columns
- Five Number Summary: Q1, Q3 and IQR calculated for TotalPrice to understand data distribution
- Outlier Detection: IQR Method applied to identify anomalous order values
- PivotTables: Used to aggregate revenue by Product, Month and Referral Source
- Conditional Formatting: Applied to visually highlight outlier records
- Charts: Bar, Line and Pie charts created to visualize key patterns

Key Findings

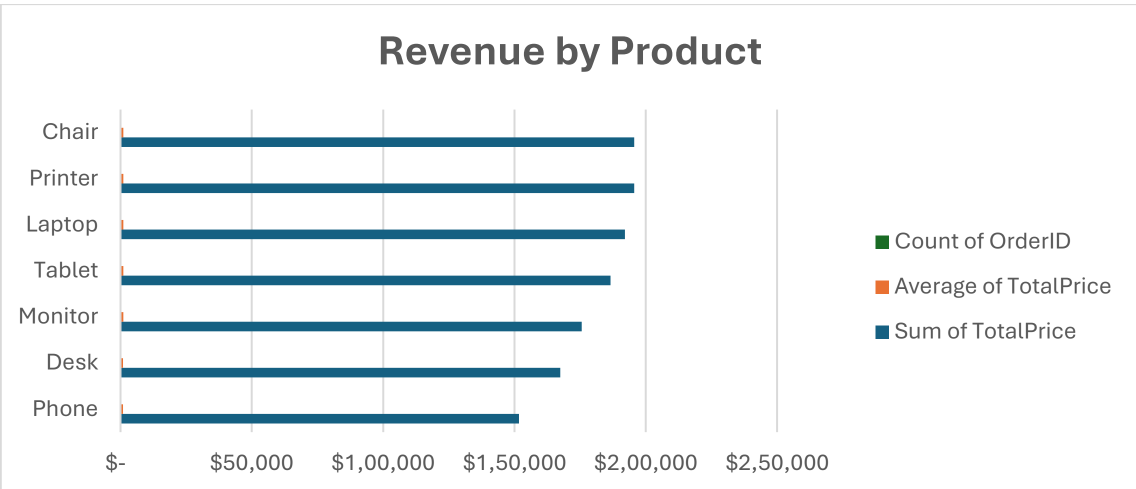
Revenue & Business Performance

1. Total Revenue: $1,264,761.96 across 1,200 orders
2. Average Order Value: $1,053.97
3. Median Order Value: $823.62
4. Revenue declined 13% from $552,643 (2023) to $480,236 (2024) — requires urgent investigation
5. June is the peak revenue month ($170,616) — nearly 2x higher than September ($69,322)



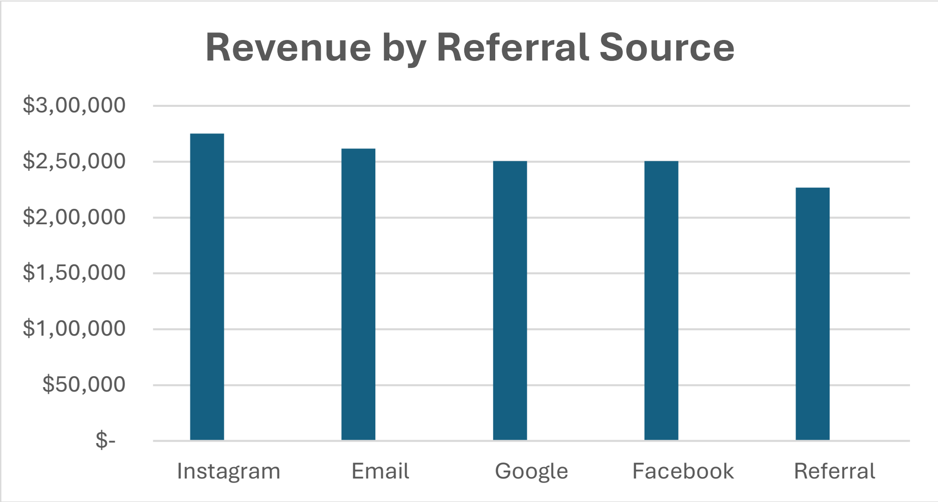
Product Performance

1. Chair is the top product by revenue ($195,620) and units sold
2. Phone is the lowest revenue product ($151,722)
3. All 7 products are evenly distributed (13%–15% each) suggesting no single dominant product



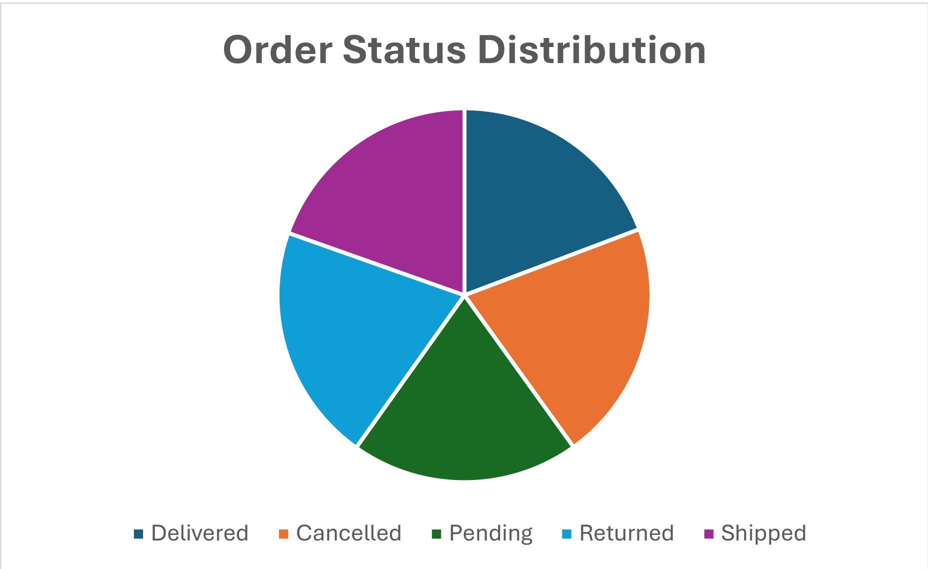
Marketing & Referrals

1. Instagram drives the most revenue ($275,285) — top performing marketing channel
2. Referral program generates the least revenue ($226,816) — needs improvement
3. All 5 referral sources are relatively balanced



Operational Issues

1. 20.8% of orders are Cancelled (250 orders)
2. 20.6% of orders are Returned (247 orders)
3. Combined Cancelled + Returned = 41.4% of all orders — a critical business problem
4. Only 19.2% of orders are successfully Delivered



Outliers

1. 8 high value orders above $3,330.41 identified using the IQR Method
2. These are SIGNAL outliers representing VIP customers — NOT data errors
3. Highest single order value: $3,456.40

Correlations

1. UnitPrice vs TotalPrice: 0.717 (Strong Positive) — higher priced items drive revenue
2. Quantity vs TotalPrice: 0.615 (Strong Positive) — bulk orders drive revenue
3. ItemsInCart vs Quantity: 0.650 (Strong Positive) — browsing behavior predicts purchasing

Recommendations

1. **Investigate Revenue Decline**

Revenue fell 13% from 2023 to 2024. Identify root cause — pricing changes, competition or reduced marketing spend

2. **Reduce Cancellation Rate**

- 20.8% cancellation rate is critically high.
- Improve checkout experience, payment options and order confirmation process

3. **Reduce Return Rate**

- 20.6% return rate suggests product descriptions or quality may not match customer expectations.
- Improve product pages and quality control

4. **Increase Instagram Investment**

- Instagram is the top revenue channel at $275,285.
- Increase marketing budget allocation here

5. **Create VIP Customer Program**

8 high value customers identified spending above $3,330 per order. Create exclusive loyalty rewards to retain them

6. **Address September Dip**

Revenue drops significantly in September ($69,322 vs $170,616 in June). Plan targeted promotions and discounts for low performing months

Conclusion

This EDA reveals that while the business generates over $1.26M in revenue, critical operational issues including a 41.4% combined cancellation and return rate are significantly impacting profitability. Immediate action on reducing order failures and capitalizing on top performing channels like Instagram will be key to reversing the 2024 revenue decline and driving sustainable growth.